Exercise 9

A jump onto the ledge

At the end, `Space` makes the walker jump, high enough to land on the ledge from the floor.

Before you start

- Exercise 8 done and pushed: the yellow sensor at the walker's feet, and `isGrounded` in the script. In a new session, run `git pull` first

1 Run 3: the jump

A key press exists only in the frame it happens, so `Update` catches it in a flag and `FixedUpdate`, where physics runs, uses it. The flag is cleared every step, so an old press can never fire later.

In `Update`, after the facing block, add:

```
        if (jumpAction.WasPressedThisFrame())
        {
            jumpRequested = true;
        }
```

In `FixedUpdate`, after the velocity line, add the jump, with the clear as the method's last line:

```
        if (jumpRequested && isGrounded)
        {
            rb.linearVelocity = new Vector2(rb.linearVelocity.x, jumpVelocity);
        }

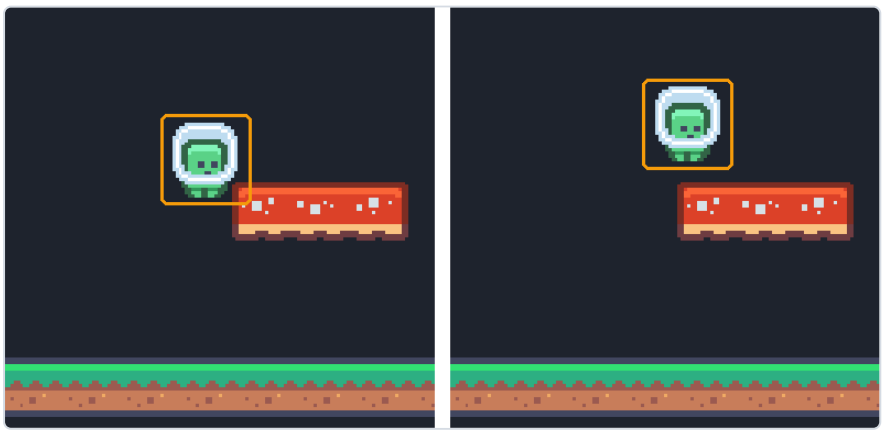
        jumpRequested = false;
```

- **Play** and check four things: `Space` jumps; a jump while running keeps the speed; `Space` in the air does nothing and the landing doesn't bounce; holding `Space` gives one jump.

2 Run 4: tune by arithmetic

A jump at speed v rises $v^2 / 2g$, and with Gravity Scale 4, $g = 4 \times 9.81 = 39.24$. The tiles, one unit each, measure the result.

- Velocity 15 rises $15^2 / (2 \times 39.24) \approx 2.9$ units: from the floor, the walker can't get onto the ledge, 3 units up. Try it.
- For 3.5 units, $v = \sqrt{(2 \times 39.24 \times 3.5)} \approx 16.6$. Stop Play, since changes made while playing are lost, then set Jump Velocity to `16.6` and try again: the walker lands on the ledge.



Velocity 15 stops short of the ledge's top (left); 16.6 clears it (right).

- For the feel, try Gravity Scale `1` with velocity `7.7`, then `6` with `18.8`: the same 3 units, one floaty, one snappy. Go back to `4` and `16.6`.
- `Ctrl` + `S`, then commit and push with the message `Add a jump that reaches the ledge`.